

Data Structures – Lab 1

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Task 1:

Source Code:

```
#include <iostream>
using namespace std;

void average(int (&tokens)[50], int n){
    float total = 0;
    for(int i = 0; i < n; i++)
    {
        total += tokens[i];
    }
    cout << "Average Token: " << total/n << endl;
}

void highest(int (&tokens)[50], int n){
    int highest = 0;
    for(int i = 0; i < n; i++)
    {
        if(highest < tokens[i])
        {
            highest = tokens[i];
        }
    }
    cout << "Highest Token:" << highest << endl;
}

void lowest(int (&tokens)[50], int n){
    int lowest = 32000;
    for(int i = 0; i < n; i++)
    {
        if(tokens[i] < lowest)
        {
            lowest = tokens[i];
        }
    }
    cout << "Lowest Token:" << lowest << endl;
}
```

```

void check(int (&tokens)[50], int n, int input)
{
    int count = 0;
    for(int i = 0; i < n; i++)
    {
        if(input == tokens[i])
        {
            count++;
        }
    }
    if(count) cout << "Yes\n";
    else cout << "No\n";
}

void reverse(int (&tokens)[50], int n){
    int first = 0, last = n-1; //Studied Dutch flag algorithm and Two
pointers in summer :)
    int temp;
    while(first < last)
    {
        temp = tokens[first];
        tokens[first] = tokens[last];
        tokens[last] = temp;
        first++;last--;
    }
    cout << "Tokens in reverse order: ";
    for(int i = 0; i < n; i++)
    {
        cout << tokens[i] << " ";
    }
    cout << endl;
}

int main(){

    int tokens[50];
    int Norders;
    cout << "Enter number of Tokens: ";
    cin >> Norders;
    cout << "Enter " << Norders << " Token numbers:\n";

    for (int i = 0; i < Norders; i++) {cin >> tokens[i];}

```

```

    average(tokens, Norders);
    highest(tokens, Norders);
    lowest(tokens, Norders);
    int input;
    cout << "Check if same token exists: ";
    cin >> input;
    check(tokens, Norders, input);
    reverse(tokens, Norders);
    return 0;
}

```

Output:

```

muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ ./a
Enter number of Tokens: 5
Enter 5 Token numbers:
2
3
4
5
6
Average Token: 4
Highest Token:6
Lowest Token:2
Check if same token exists: 4
Yes
Tokens in reverse order: 6 5 4 3 2
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$

```

Task 2:

Source Code:

```

#include <iostream>
#include <string>
using namespace std;

```

```

int main(){
    string rollNo;
    cout << "Enter Roll no: ";
    cin >> rollNo;
    int len = rollNo.length();
    if(len == 8)cout << len << " is the length of Roll No. hence
Validated!\n";
    else cout << len << " is the length of Roll No. hence Length is
Invalid!\n";
    int Hindex = rollNo.find('-');
    if(Hindex==3) cout << "Valid Hyphen index.\n";
    else cout << "Invalid Hyphen Index!\n";
    string before_hyphen = rollNo.substr(0, Hindex);
    string after_hyphen = rollNo.substr(Hindex+1, len-1);
    cout << "Expected string before Hyphen: " << before_hyphen << endl
    << "Expected string after Hyphen: " << after_hyphen << endl;

    string email;
    cout << "Enter NU email: ";
    cin >> email;
    int eIdent = email.find('@');
    if(eIdent > 0) cout << "Valid email\n";
    else cout << "Missing @ symbol, Invalid email!\n";
    int domain = email.find("nu.edu.pk");
    if(domain > 0) cout << "Valid Domain!\n";
    else cout << "Invalid Domain!\n";
    string ErollNo = email.substr(0, eIdent);
    cout << "Roll No extracted from email: " << ErollNo << endl;
    string rollNo2;
    cout << "Input another roll no: ";
    cin >> rollNo2;
    int comp = rollNo.compare(rollNo2);
    if(!comp) cout << "Match!\n";
    else cout << "Not Match!\n";
    string count = before_hyphen.substr(0, Hindex-1) + after_hyphen;
    cout << "Digit count in roll no: " << count.length() << endl;
    if(count.length() == 6) cout << "Exactly 6 digits are present!\n";
}

```

Output:

```
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ g++ Task2.cpp -o a
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ ./a
Enter Roll no: 25I-3207
8 is the length of Roll No. hence Validated!
Valid Hyphen index.
Expected string before Hyphen: 25I
Expected string after Hyphen: 3207
Enter NU email: i253207@isb.nu.edu.pk
Valid email
Valid Domain!
Roll No extracted from email: i253207
Input another roll no: 25P-0011
Not Match!
Digit count in roll no: 6
Exactly 6 digits are present!
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ |
```

Task 3:

Source Code:

```
#include <iostream>
using namespace std;

void bugTransferFee(double from, double to, double fee){
    to += fee;
    from -= fee;
    cout << "\nAmount Transferred via buggy function\n" << endl;
}

/*bugTransferFee didn't work because it made changes
to the variables with scope only to the void function
instead of the actual variables in main function
This problem is faced when argument is passed by value
instead of pass of reference*/
void transferFee(double &from, double &to, double &fee){
    to += fee;
    from -= fee;
```

```

    cout << "\nAmount Transferred via reference function\n" << endl;
}

void pointerTransferFee(double *from, double *to, double *fee){
    *to += *fee;
    *from -= *fee;
    cout << "\nAmount Transferred via Pointer function\n" << endl;
}

void swapBalances(double *a, double *b){
    *a += *b;
    *b = *a-(*b);
    *a -= *b;
    cout << "\nAmount swapped via Swap Balance function\n" << endl;
}

int main(){
    double Fee, feeFrom, feeTo;
    cout << "Enter Account 1 amount: ";
    cin >> feeFrom;
    cout << "Enter fee to be transferred: ";
    cin >> Fee;
    cout << "Enter the Account 2 current amount to transfer fee to: ";
    cin >> feeTo;
    cout << "***Original Balances*** \n" << "Account 1: " << feeFrom <<
endl
    << "Account 2: " << feeTo << endl;
    bugTransferFee(feeFrom, feeTo, Fee);
    cout << "***After Buggy Function*** \n" << "Account 1: " << feeFrom
<< endl
    << "Account 2: " << feeTo << endl;
    transferFee(feeFrom, feeTo, Fee);
    cout << "***After Reference Function*** \n" << "Account 1: " <<
feeFrom << endl
    << "Account 2: " << feeTo << endl;
    pointerTransferFee(&feeFrom, &feeTo, &Fee);
    cout << "***After Reference Function*** \n" << "Account 1: " <<
feeFrom << endl
    << "Account 2: " << feeTo << endl;
    swapBalances(&feeFrom, &feeTo);

```

```

        cout << "***After Swap Balance Function*** \n" << "Account 1: " <<
feeFrom << endl
        << "Account 2: " << feeTo << endl;
}

```

Output:

```

muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ g++ Task3.cpp -o a
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ ./a
Enter Account 1 amount: 10000
Enter fee to be transferred: 500
Enter the Account 2 current amount to transfer fee to: 2000
***Original Balances***
Account 1: 10000
Account 2: 2000

Amount Transferred via buggy function

***After Buggy Function***
Account 1: 10000
Account 2: 2000

Amount Transferred via reference function

***After Reference Function***
Account 1: 9500
Account 2: 2500

Amount Transferred via Pointer function

***After Reference Function***
Account 1: 9000
Account 2: 3000

Amount swapped via Swap Balance function

***After Swap Balance Function***
Account 1: 3000
Account 2: 9000
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ |

```

Source Code:

```

#include <iostream>
using namespace std;

int add(int a, int b){return a + b;}

int subtract(int a, int b){return a - b;}

int multiply(int a, int b){return a * b;}

int divide(int a, int b){return a / b;}

int compute(int (*operate)(int, int), int x, int y){return operate(x,
y);}

int main(){

```



```

int (*calculator[4])(int, int) = {add, subtract, multiply, divide};
cout << "\n=== Calculator ===\n";
cout << "1. Add\n";
cout << "2. Subtract\n";
cout << "3. Multiply\n";
cout << "4. Divide\n";
int choice, x, y;
cout << "Enter your choice: ";
cin >> choice;
cout << "Enter first int: ";
cin >> x;
cout << "Enter second int: ";
cin >> y;
int (*operation)(int, int) = calculator[choice-1];
int result = operation(x, y);
cout << "Result: " << result << endl;
cout << "\n=== Direct Callback Tests ===\n";
cout << "compute(add, 1, 2) = " << compute(add, 1, 2) << endl;
cout << "compute(subtract, 14, 10) = " << compute(subtract, 14, 10)
<< endl;
    cout << "compute(multiply, 2, 3) = " << compute(multiply, 2, 3) <<
endl;
    cout << "compute(divide, 6, 3) = " << compute(divide, 6, 3) << endl;
    return 0;
}

```

Output:

```
muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ g++ Task4.cpp -o a
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ ./a

=== Calculator ===
1. Add
2. Subtract
3. Multiply
4. Divide
Enter your choice: 1
Enter first int: 1
Enter second int: 2
Result: 3

=== Direct Callback Tests ===
compute(add, 1, 2) = 3
compute(subtract, 14, 10) = 4
compute(multiply, 2, 3) = 6
compute(divide, 6, 3) = 2
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$
```

Task 5:

Source Code:

```
#include <iostream>
#include <string>
using namespace std;

struct Student { \
    int rollNo;
    string name;
    float gpa;
};

void displayAll(Student arr[], int n){
    cout << "\n===Student Records===\n";
    cout << "Roll No.   Name   GPA\n";
    for(int i = 0; i < n; i++){
        cout << arr[i].rollNo << "   " << arr[i].name << "   " << arr[i].gpa << "\n";
    }
}
```

```

        cout << arr[i].rollNo << "      " << arr[i].name
        << "      " << arr[i].gpa << endl;
    }
}

Student* findTopper(Student arr[], int n){
    float top = 0.0;
    Student* ptr = nullptr;
    for(int i = 0; i < n; i++)
    {
        if(top < arr[i].gpa)
        {
            top = arr[i].gpa;
            ptr = &arr[i];
        }
    }
    return ptr;
}

void updateGPA(Student *s, float newGpa){s->gpa = newGpa;}

int main(){
    Student batch[10];
    cout << "Enter number of students: ";
    int num;
    cin >> num;
    for (int i = 0; i < num; i++)
    {
        cout << "Enter details of Student " << i+1 << ": \n";
        cout << "Roll No: "; cin >> batch[i].rollNo;
        cout << "Name: "; cin >> batch[i].name;
        cout << "GPA: "; cin >> batch[i].gpa;
    }
    displayAll(batch, num);
    Student* topper = findTopper(batch, num);
    cout << "\n_____\\n===Topper Student===\\n";
    cout << "Name: " << topper->name << "\\nGPA: " << topper->gpa <<
endl;
    cout << "\n_____\\n===GPA Update===\\n";
    cout << "Enter roll no of student to update GPA: ";

```

```

int input;float newGPA;
cin >> input;
cout << "Enter new GPA: "; cin >> newGPA;
for(int i = 0; i < num; i++)
{
    if(input == batch[i].rollNo)
    {
        cout << "Original GPA of " << batch[i].name << ": " <<
batch[i].gpa << endl;
        updateGPA(&batch[i], newGPA);
        cout << "Updated GPA of " << batch[i].name << ": " << batch[i].gpa
<< endl;
        break;
    }
}
}

```

Output:

```

muhammad-awais@muhammad-awais: ~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1
~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ g++ Task5.cpp -o a
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$ ./a
Enter number of students: 5
Enter details of Student 1:
Roll No: 1
Name: Ali
GPA: 3.0
Enter details of Student 2:
Roll No: 2
Name: Raza
GPA: 3.8
Enter details of Student 3:
Roll No: 3
Name: Taha
GPA: 2.5
Enter details of Student 4:
Roll No: 4
Name: Daniyal
GPA: 4.0
Enter details of Student 5:
Roll No: 5
Name: Awais
GPA: 2.6

===Student Records===
Roll No.   Name   GPA
-----
1         Ali    3
2         Raza   3.8
3         Taha   2.5
4         Daniyal 4
5         Awais  2.6

===Topper Student===
Name: Daniyal
GPA: 4

===GPA Update===
Enter roll no of student to update GPA: 5
Enter new GPA: 3.7
Original GPA of Awais: 2.6
Updated GPA of Awais: 3.7
muhammad-awais@muhammad-awais:~/VM/Shared/FAST/Semester3/Assignments/DSA/Lab/Lab1$

```